

Lab - AzCopy Tool

To Begin with the lab

Summary

In this lab, you learn how to use the **AzCopy** command-line utility to transfer data between a local machine and an Azure Storage Account. You install the AzCopy tool, create a Blob Container, and upload a file to Azure Blob Storage using a **Shared Access Signature (SAS)** token. AzCopy provides a fast and efficient way to manage Azure Storage resources from the command line.

Understanding

AzCopy is a Microsoft command-line tool used to copy files and folders between local storage and Azure Storage services such as **Blob Storage** and **Azure Files**. It is available for **Windows, Linux, and macOS**.

Unlike Azure Storage Explorer, which provides a graphical interface, AzCopy is designed for developers and administrators who need to automate storage operations or transfer large amounts of data efficiently.

AzCopy uses a **Shared Access Signature (SAS)** token to authenticate requests without exposing the Storage Account access keys. With AzCopy, you can perform operations such as creating containers, uploading blobs, downloading blobs, synchronizing data, and copying files between storage accounts.

Example

Suppose you have a local SQL script named **init.sql** that needs to be uploaded to Azure Blob Storage.

Local File:

D:\Dev3\init.sql

Storage Account:

stdeveastus10

Blob Container:

scripts

Blob:

init.sql

Application Flow:

Local Machine



AzCopy Tool



Azure Storage Account



Blob Container (scripts)



init.sql (Blob)

The **init.sql** file is uploaded to the **scripts** container and becomes available in Azure Blob Storage.

Lab Steps

Step 1: Download AzCopy

- Open the AzCopy download page.
- Download the appropriate version for your operating system.
- For Windows, download the **Windows 64-bit ZIP** package.

Get AzCopy

If you're using AzCopy on a Linux machine, you can use a package manager. For all other operating systems, download a portable binary file. For detailed information on AzCopy releases, see the [AzCopy release page](#).

Use a package manager (Linux only)

Installing AzCopy through your Linux distribution's package manager is the most convenient and maintainable approach. Package manager installation provides automatic dependency resolution, simplified updates, and integration with your system's software management. For step-by-step guidance, see [Install AzCopy on Linux by using a package manager](#).

Download a portable binary

An installation package is available only for Linux. For all other operating systems, download the AzCopy v10 executable file to any directory on your computer.

- **Windows 64-bit (zip)**
- Windows 32-bit (zip)
- Windows ARM64 (zip)
- Linux x86-64 (tar)
- Linux ARM64 (tar)
- macOS (zip)
- macOS ARM64 (zip)

- Extract the ZIP file.
- Copy **AzCopy.exe** to a convenient folder (for example, **D:\Dev3**).

Step 2: Prepare the Local File

- Place the file you want to upload in the same directory as **AzCopy.exe**.

```
PS D:\Dev3> ls

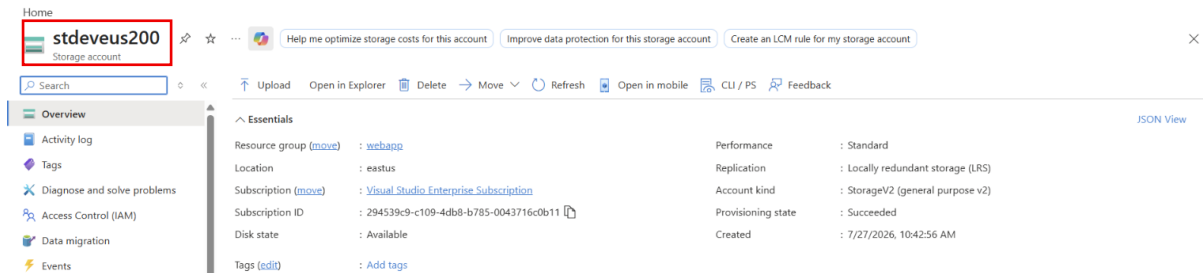
Directory: D:\Dev3

Mode                LastWriteTime         Length Name
----                -
-a-----         27-07-2026      15:43      60792672 azcopy.exe

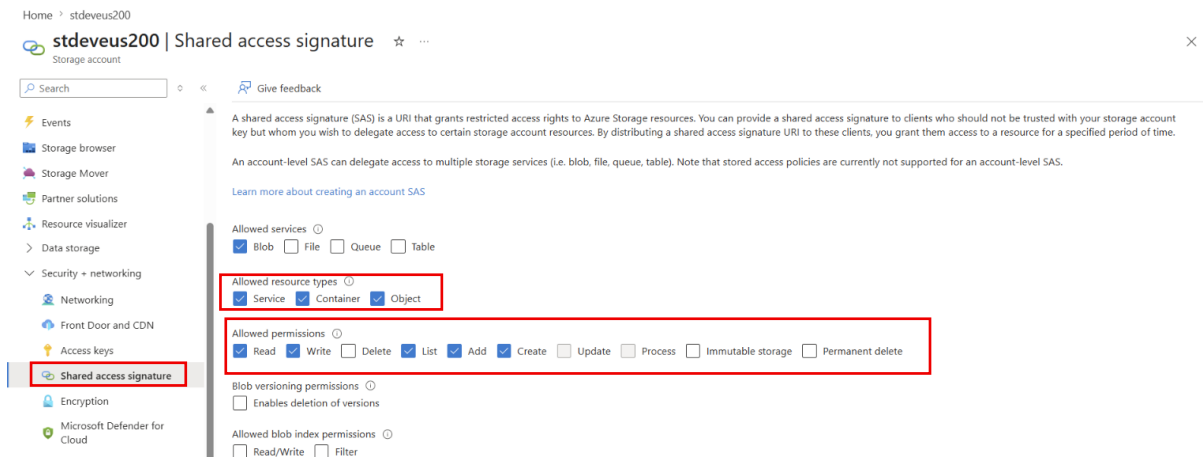
PS D:\Dev3>
```

Step 3: Generate a SAS Token

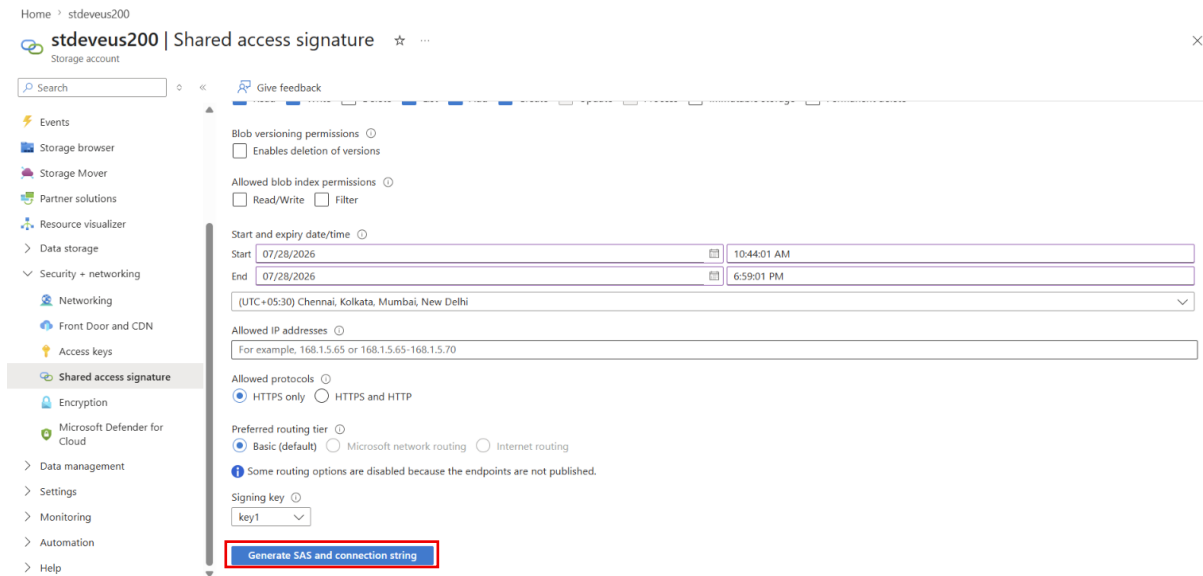
- Open the **Azure Portal**.
- Navigate to your **Storage Account**.



- Select **Shared access signature**.
- Select **Blob** as the allowed service.
- Select the required **resource types**.
- Choose the **required permissions (Read, Write, Create, List, etc.)**.



- Click **Generate SAS and connection string**.



- Copy the generated **SAS Token**.

Home > stdeveus200

stdeveus200 | Shared access signature ☆ ...

Search

Give feedback

End 07/28/2026 6:59:01 PM

(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi

Allowed IP addresses

For example, 168.1.5.65 or 168.1.5.65-168.1.5.70

Allowed protocols

☒ HTTPS only ☐ HTTPS and HTTP

Preferred routing tier

☒ Basic (default) ☐ Microsoft network routing ☐ Internet routing

Some routing options are disabled because the endpoints are not published.

Signing key

key1

Generate SAS and connection string

Connection string

BlobEndpoint=https://stdeveus200.blob.core.windows.net/QueueEndpoint=https://stdeveus200.queue.core.windows.net/FileEndpoint=https://stdeveus200.file.core.windows.net/TableEndpoint=https://st...

SAS token

sv=2026-02-06&ss=b&srt=sco&sp=rlac&se=2026-07-28T13:29:01Z&st=2026-07-28T05:14:01Z&spr=https&sig=d3jLcEffEhCTpTZaUfcT0598n8Jl021m3uNIWBQ1%2BYk%3D

Blob service SAS URL

https://stdeveus200.blob.core.windows.net/?sv=2026-02-06&ss=b&srt=sco&sp=rlac&se=2026-07-28T13:29:01Z&st=2026-07-28T05:14:01Z&spr=https&sig=d3jLcEffEhCTpTZaUfcT0598n8Jl021m3uNIWB...

Step 4: Create a Blob Container Using AzCopy

Run the following command:

```
azcopy make "https://<storage-account>.blob.core.windows.net/scripts?<SAS-Token>"
```

Example:

```
azcopy make "https://stdeveus200.blob.core.windows.net/scripts? sv=2026-02-06&ss=b&srt=sco&sp=rlac&se=2026-07-28T13:29:01Z&st=2026-07-28T05:14:01Z&spr=https&sig=d3jLcEffEhCTpTZaUfcT0598n8Jl021m3uNIWBQ1%2BYk%3D"
```

- Creates a new Blob Container.
- Uses the SAS Token for authentication.

```
Windows PowerShell
PS D:\Dev3> .\azcopy make "https://stdeveus200.blob.core.windows.net/scripts?sv=2026-02-06&ss=b&srt=sco&sp=rlac&se=2026-07-28T13:29:01Z&st=2026-07-28T05:14:01Z&spr=https&sig=d3jLcEffEhCTpTZaUfcT0598n8Jl021m3uNIWBQ1%2BYk%3D"

Successfully created the resource.
PS D:\Dev3> |
```

Step 5: Verify the Container

- Open the Azure Portal.
- Navigate to **Storage Account** → **Containers**.
- Refresh the page.
- Verify that the **scripts** container has been created.

Home > stdeveus200

stdeveus200 | Containers ☆ ...

Search

+ Add container Upload Refresh Delete Change access level Restore containers Edit columns

Search containers by prefix

Only show active containers

Showing all 2 items

	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	Scripts	7/27/2026, 10:43:15 AM	Private	Available	-
☒	scripts	7/28/2026, 11:14:58 AM	Private	Available	-

Step 6: Upload a Blob Using AzCopy

Run the following command:

```
azcopy copy " " "https://<storage-account>.blob.core.windows.net/scripts/init.sql?<SAS-Token>"
```

Example:

```
azcopy copy "D:\Dev3\init.sql"
"https://stdeveus200.blob.core.windows.net/scripts/init.sqlsv=2026-02-
06&ss=b&srt=sco&sp=rwlac&se=2026-07-28T13:29:01Z&st=2026-07-
28T05:14:01Z&spr=https&sig=d3jLcEffEhCTpTZaUfcT0598n8Jl021m3uNIWBQ1%2BYk%3D"
```

- Uploads the local file.
- Stores it in the **scripts** Blob Container.

```
Mode                LastWriteTime         Length Name
-----
-a-----          27-07-2026      15:43      60792672 azcopy.exe
-a-----          28-07-2026      11:22           0 init.sql

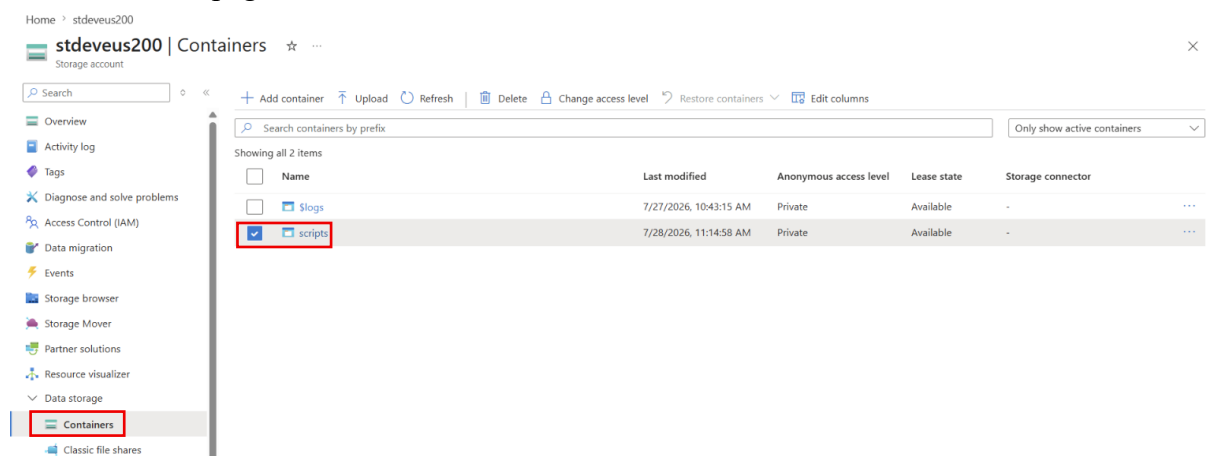
PS D:\Dev3> .\azcopy copy "D:\Dev3\init.sql" "https://stdeveus200.blob.core.windows.net/scripts/init.sql?sv=2026-02-06&ss=b&srt=sco&sp=rwlac&se=2026-07-28T13:29:01Z&st=2026-07-28T05:14:01Z&spr=https&sig=d3jLcEffEhCTpTZaUfcT0598n8Jl021m3uNIWBQ1%2BYk%3D"
INFO: Any empty folders will not be processed, because source and/or destination doesn't have full folder support
INFO: Scanning...

Job b9423717-ad29-9049-7084-4c23a3b25f99 has started
Log file is located at: C:\Users\rrath\azcopy\b9423717-ad29-9049-7084-4c23a3b25f99.log
100.0 %, 1 Done, 0 Failed, 0 Pending, 0 Skipped, 1 Total,

Job b9423717-ad29-9049-7084-4c23a3b25f99 summary
Elapsed Time (Minutes): 0.0667
Number of File Transfers: 1
Number of Folder Property Transfers: 0
Number of Symlink Transfers: 0
Total Number of Transfers: 1
Number of File Transfers Completed: 1
Number of Folder Transfers Completed: 0
Number of File Transfers Failed: 0
Number of Folder Transfers Failed: 0
Number of File Transfers Skipped: 0
Number of Folder Transfers Skipped: 0
Number of Symbolic Links Skipped: 0
Number of Hardlinks Converted: 0
Number of Hardlinks Skipped: 0
Number of Special Files Skipped: 0
Total Number of Bytes Transferred: 0
Final Job Status: Completed
```

Step 7: Verify the Uploaded Blob

- Open the Azure Portal.
- Navigate to **Storage Account** → **Containers** → **scripts**.
- Refresh the page.



- Verify that **init.sql** appears in the container.

scripts

Container

Search

+ Add Directory Upload Change access level Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Shared access tokens

Access policy

Properties

Metadata

scripts

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	init.sql	7/28/2026, 11:25:23 AM	Hot (Inferred)	Block blob	0	Available	...